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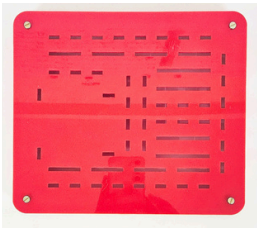
SESSION NAME

Session

3

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1


2

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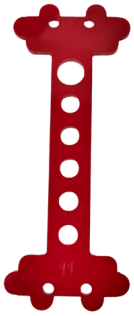


1

Spacers

2

L Clamp

3

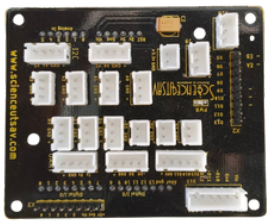
Spacer Screws

4

Nuts

5

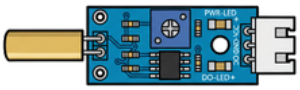
Screw

6

Arduino With Shield

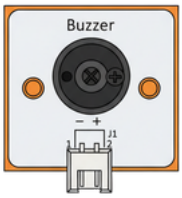
7

Arduino cable

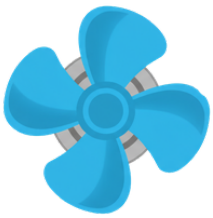
ELECTRONIC COMPONENTS



Tilt Sensor- A Tilt Sensor is a simple device used to detect the orientation or tilt of an object. It changes its output when the sensor is tilted from its normal position. It is commonly used in security systems, motion-detection projects, and automatic switching applications.



Buzzer- A buzzer is an electronic component that generates sound when electric current flows through it. It is commonly utilized in alarms, alerts, and notification systems to provide auditory signals.



DC Motor – A DC Motor is an electric motor that converts direct current (DC) electrical energy into rotational motion. It rotates continuously when power is supplied and is commonly used in toys, fans, robots, and other projects that require simple motorized movement.

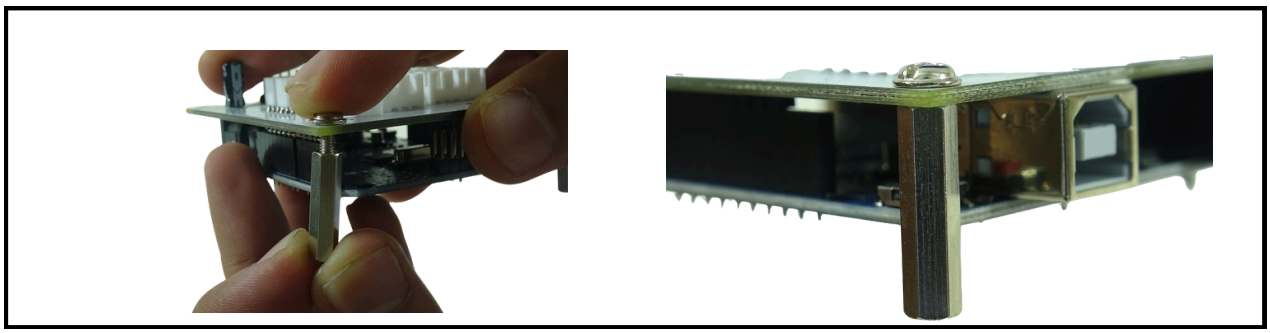
7-Segment Display– A Seven Segment Display is an electronic display device used to show numbers and a few characters using seven individual LED segments arranged in the shape of the number 8. It is commonly used in digital clocks, counters, calculators, and electronic measuring devices.

2 PIN RMC

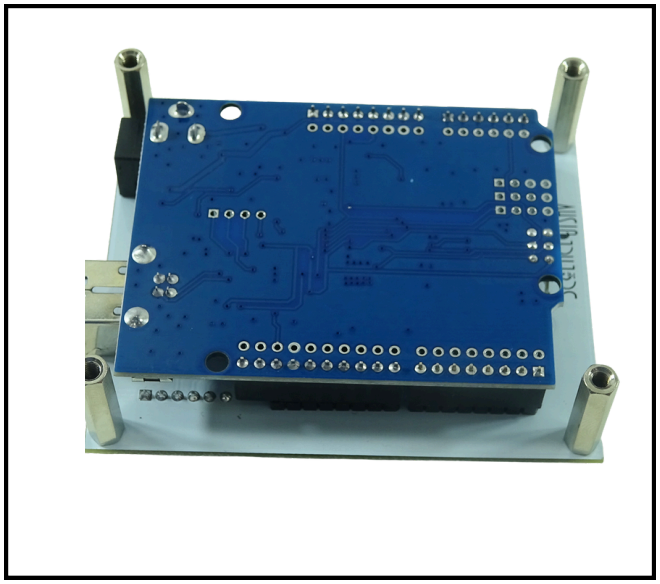
Used to connect the 1-watt LEDs and the buzzer.

Procedure :

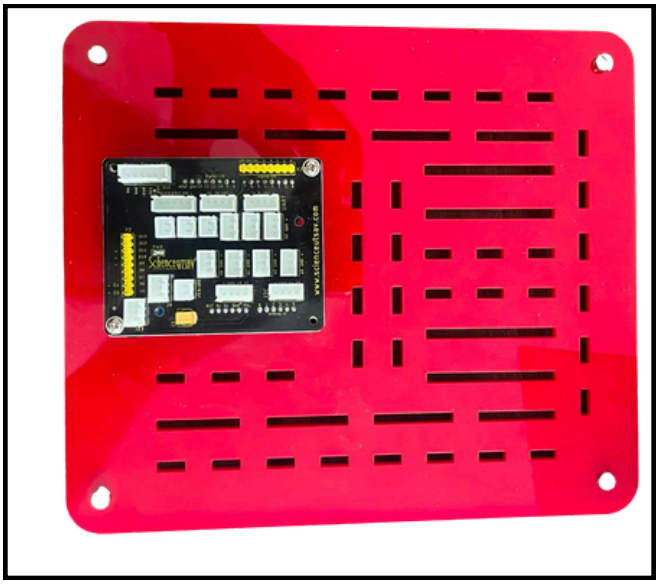
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw



Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



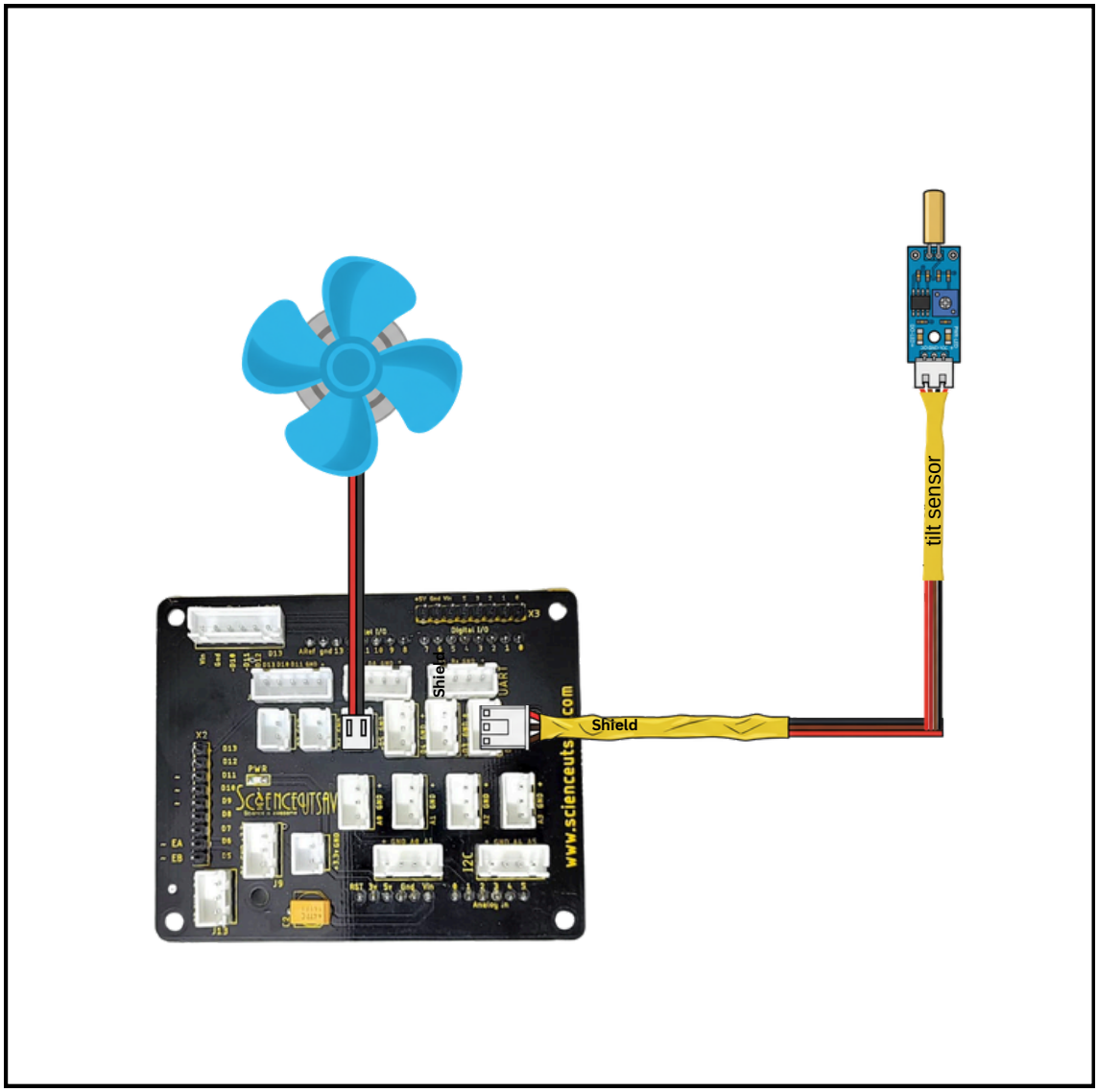
Step 3 : Place the Arduino Shield to the base board as shown below



Step 4 : Fix it with the spacer screw from the bottom as shown below

BEGINNER PROJECT

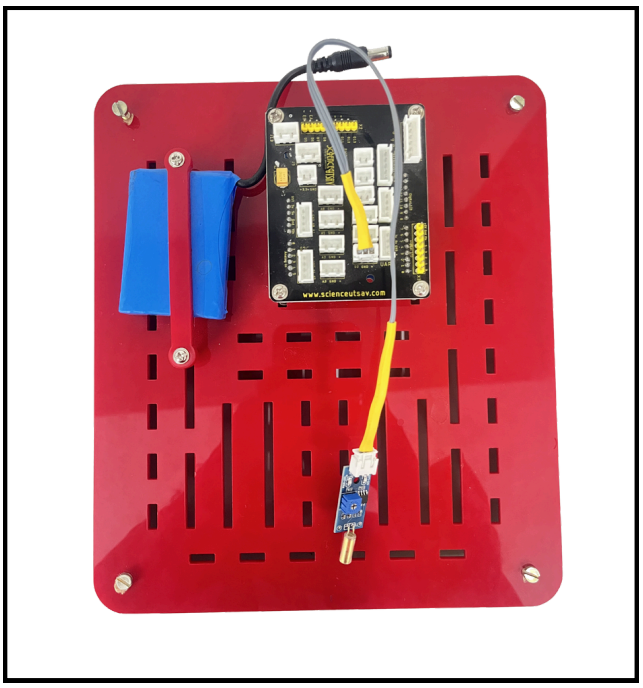
The circuit diagram of the project is given below



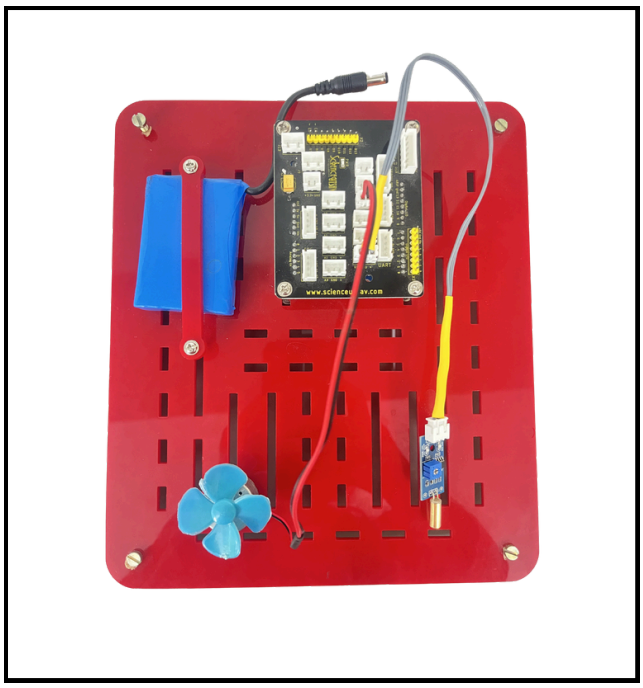
CONNECTION:

1. DC motor- D5
2. Tilt sensor- D2

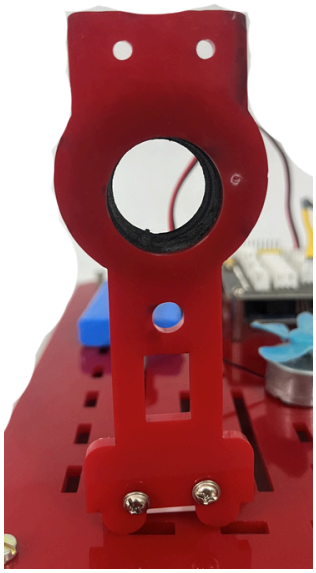
Step 1- Connect the tilt sensor to the board



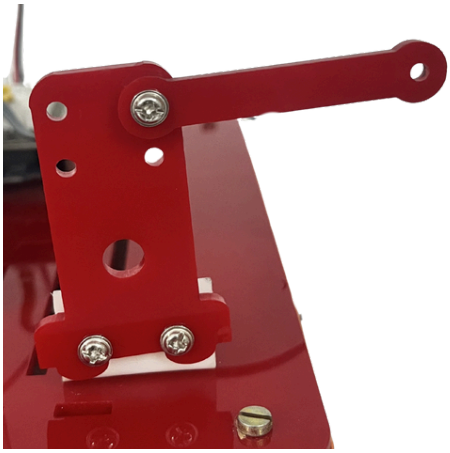
Step 2- Attach the DC motor as shown



Step 3- Attach the DC motor holder as shown

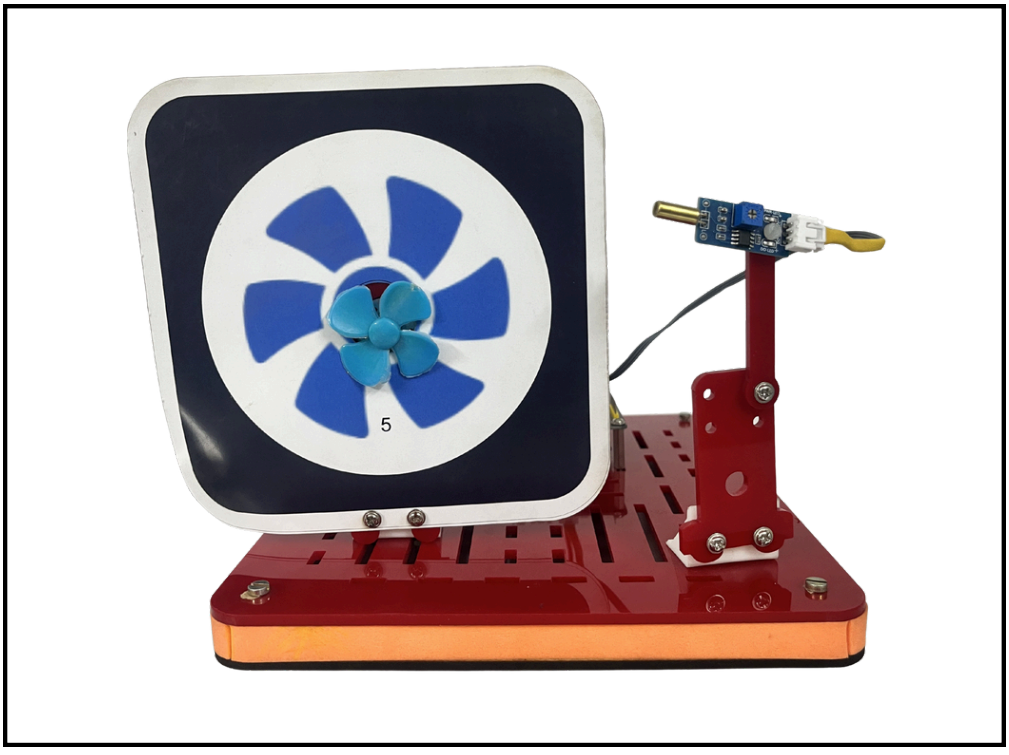


Step 4- Fix the other stand along with the other structure as shown



Step 5- Attach the template and fix the fan

Step 6- Fix the sensor to the structure as shown

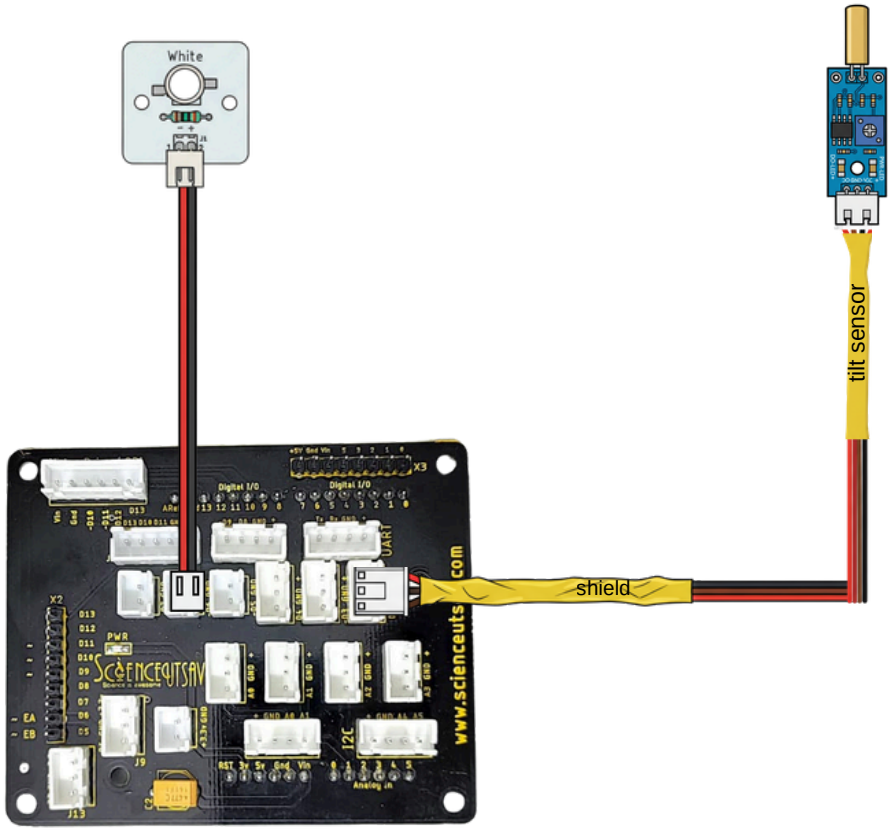


code

```
// Session 3 Grade 6
// Exhaust Fan
// Tilt sensor tilted = fan motor ON

const int tiltSensor = 2;
const int motor = 5;

const int TILT_ACTIVE = HIGH;  // change to LOW if opposite

void setup() {
  pinMode(tiltSensor, INPUT);
  pinMode(motor, OUTPUT);
}

void loop() {
  if (digitalRead(tiltSensor) == TILT_ACTIVE) {
    digitalWrite(motor, HIGH);
  } else {
    digitalWrite(motor, LOW);
  }
}
```

BEGINNER PROJECT

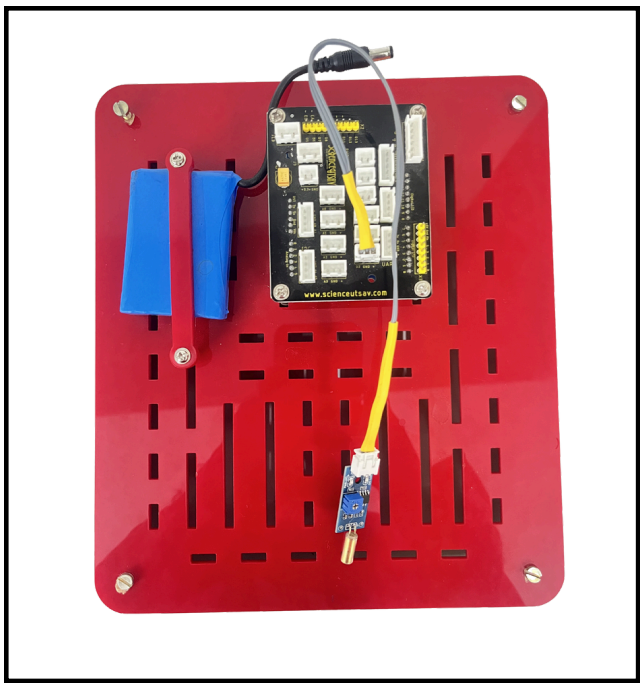
The circuit diagram of the project is given below



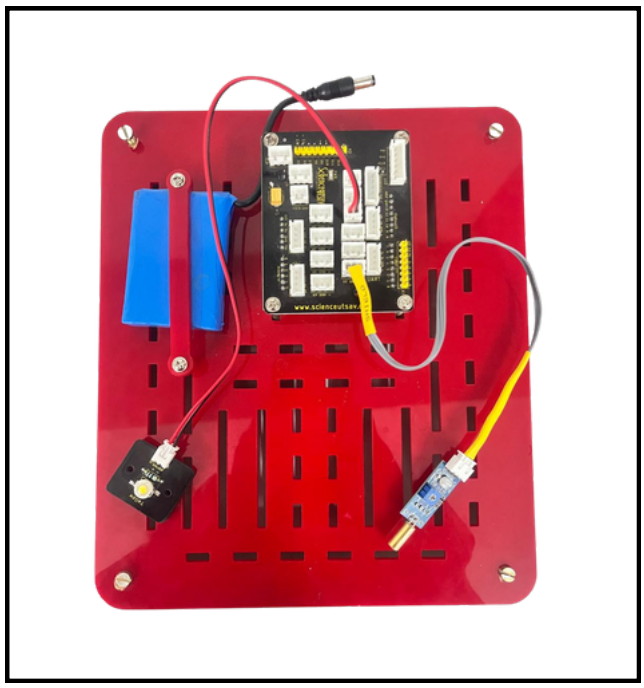
CONNECTION:

1. 1W LED- D5
2. Tilt sensor- D2

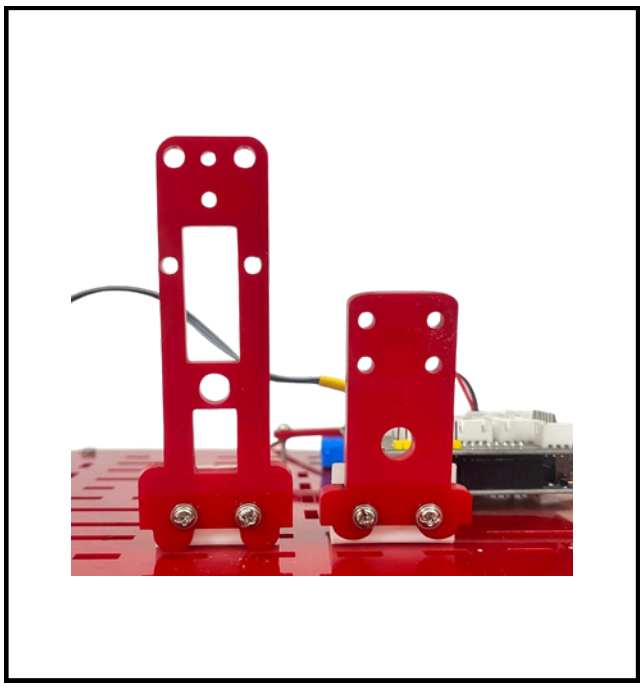
Step 1- Connect the tilt sensor to the board



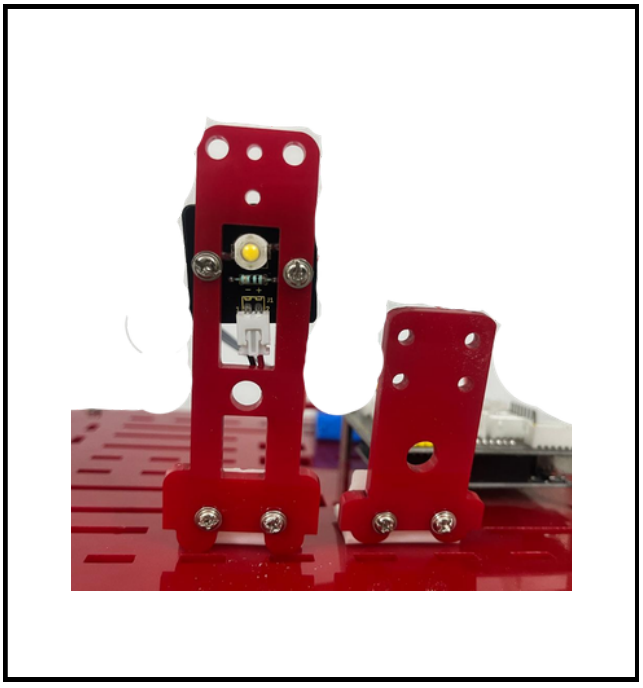
Step 2- Attach the LED as shown



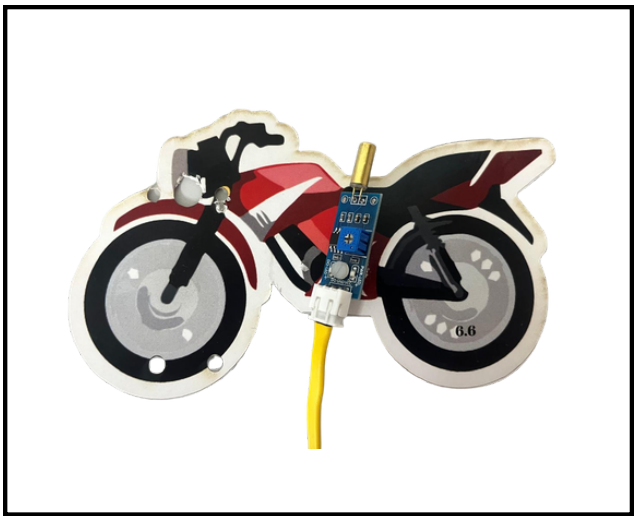
Step 3- Attach the stands to the board



Step 4- Fix the LED to the stand as shown



Step 5- Attach the sensor to the template as shown



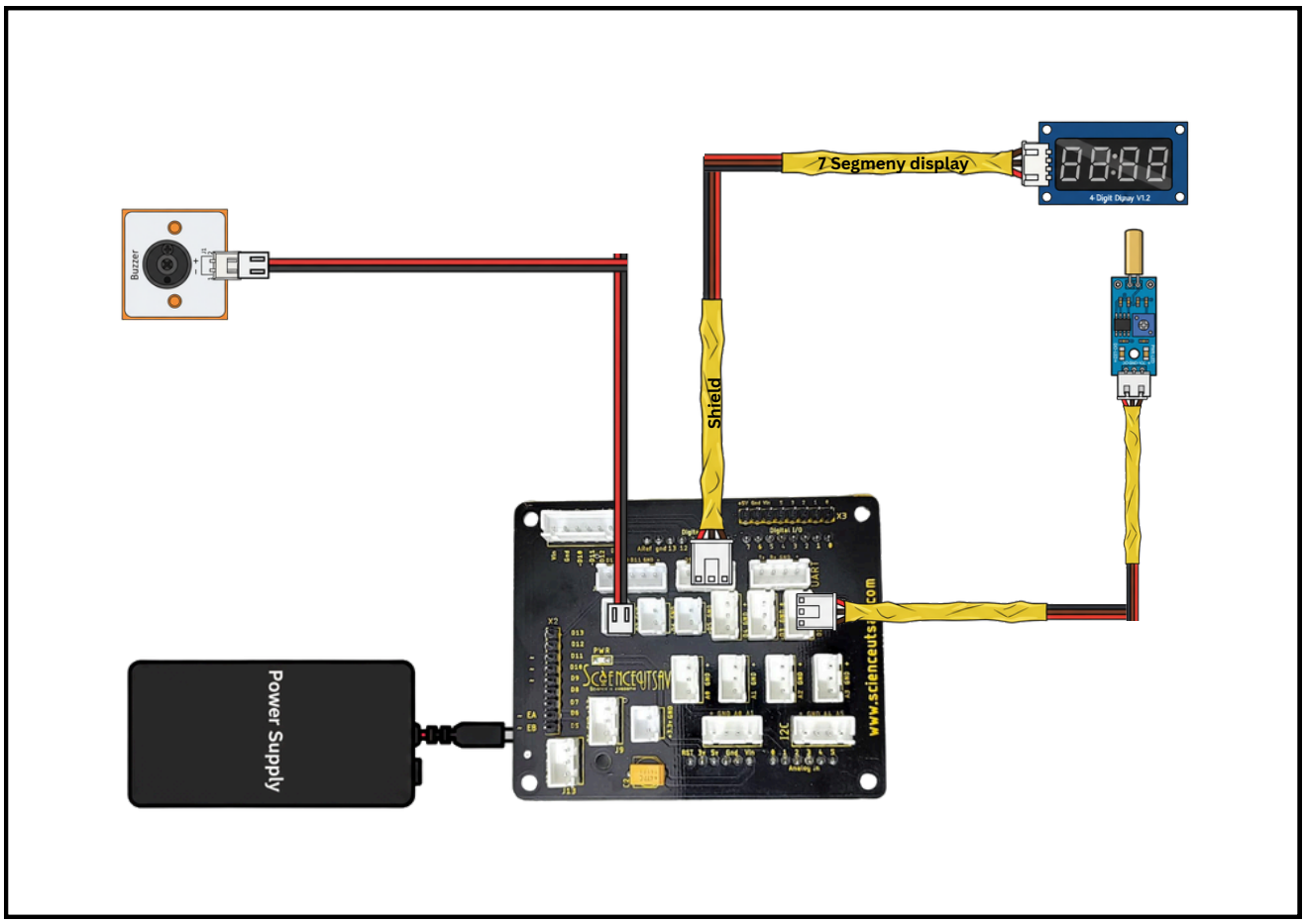
Step 6- Fix the template to the stands

code

```
//Session 3 Grade 6
// Bike Stand
// Stand open = LED OFF
// Stand released = LED ON

const int tiltSensor = 2;
const int led = 6;

const int STAND_OPEN = HIGH;  // change to LOW if opposite

void setup() {
  pinMode(tiltSensor, INPUT);
  pinMode(led, OUTPUT);
}

void loop() {
  if (digitalRead(tiltSensor) == STAND_OPEN) {
    digitalWrite(led, LOW);
  } else {
    digitalWrite(led, HIGH);
  }
}
```

INTERMEDIATE PROJECT

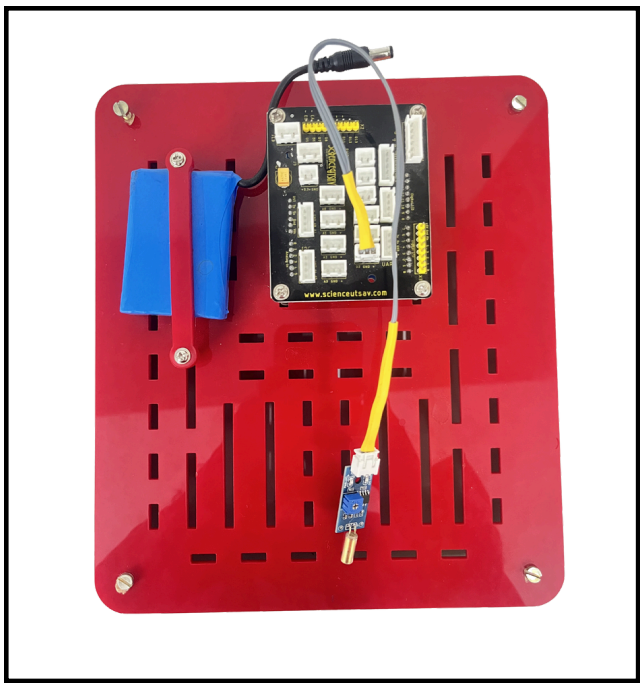
The circuit diagram of the project is given below



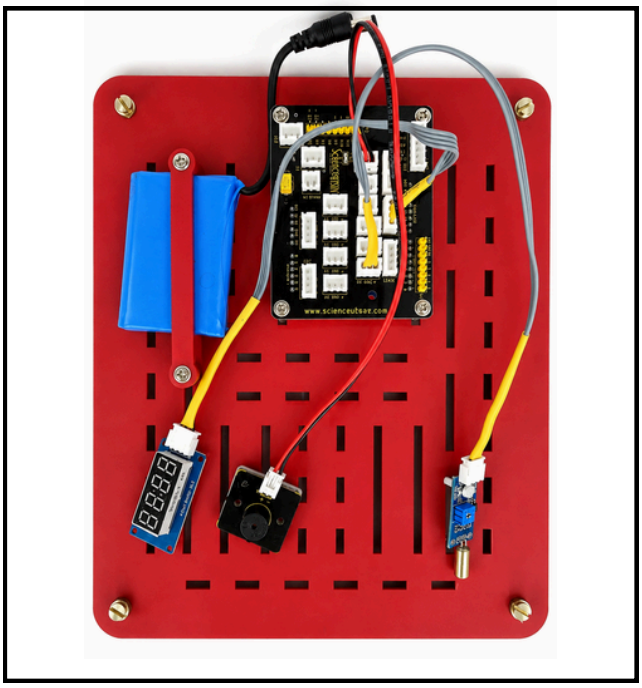
CONNECTION:

1. Tilt sensor- D2
2. 7-segment display- D9, D8
3. Buzzer- D7

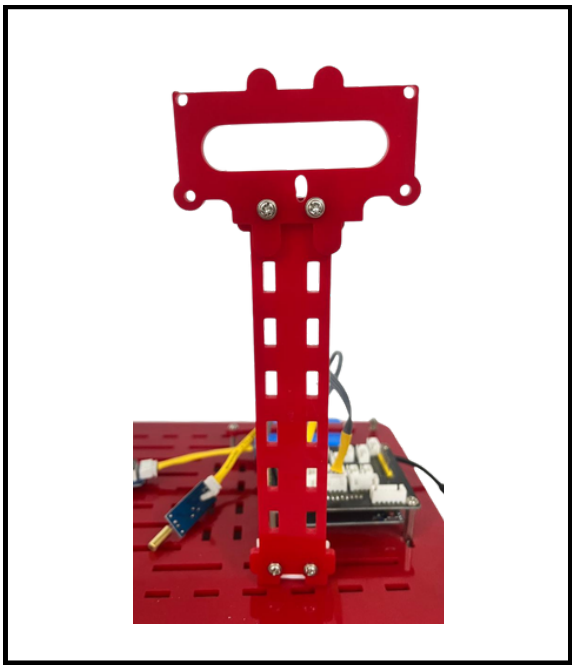
Step 1- Connect the tilt sensor to the board



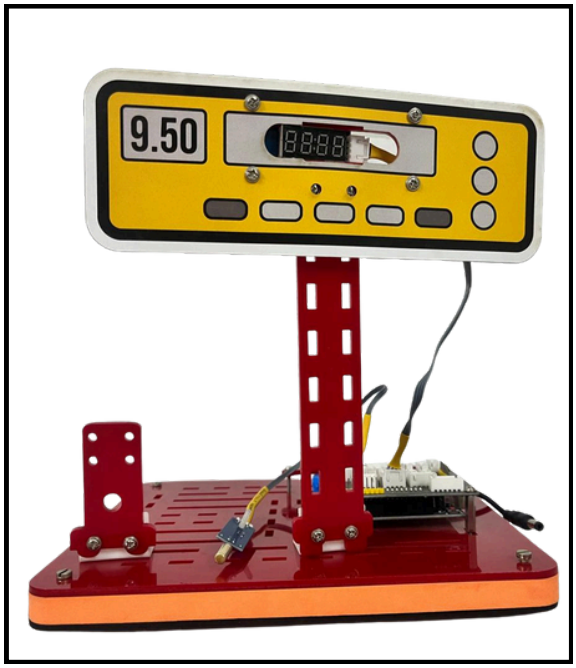
Step 2- Attach the display and buzzer



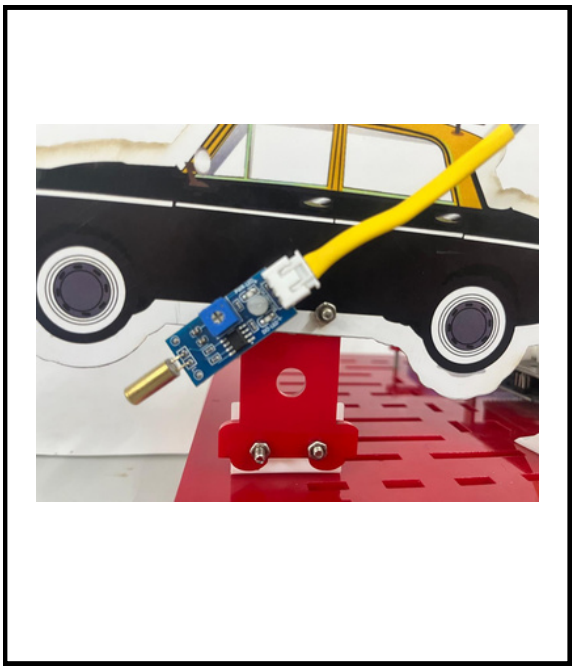
Step 3- Attach the stands as shown



Step 4- Attach the template and fix the display to it



Step 5- Attach the other template to the small stad and fix the tilt senor to it

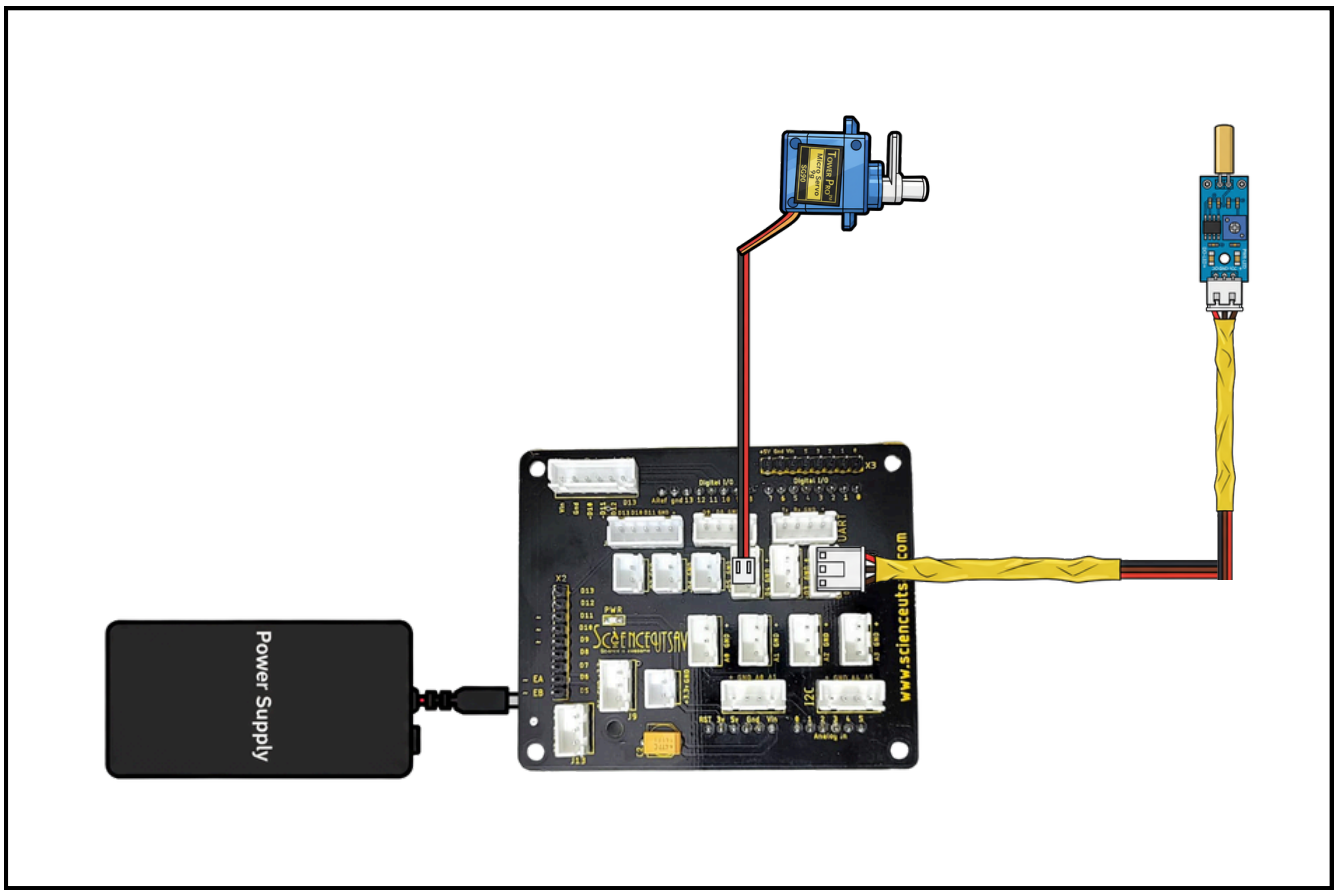


code

```
// Session 3 Grade 7
// Taxi Meter
// Tilt meter down = price starts increasing

#include <TM1637Display.h>

const int tiltSensor = 2;
const int buzzer = 7;

const int CLK = 9;
const int DIO = 8;

const int TILT_ACTIVE = HIGH;  // change to LOW if opposite

TM1637Display display(CLK, DIO);

int price = 0;
unsigned long lastUpdate = 0;

void setup() {
  pinMode(tiltSensor, INPUT);
  pinMode(buzzer, OUTPUT);

  display.setBrightness(7);
  display.showNumberDec(0, true);
}

void loop() {
  if (digitalRead(tiltSensor) == TILT_ACTIVE) {
    if (millis() - lastUpdate > 700) {
      price = price + 5;

      if (price > 9999) {
        price = 0;
      }

      display.showNumberDec(price, true);

      digitalWrite(buzzer, HIGH);
      delay(100);
      digitalWrite(buzzer, LOW);

      lastUpdate = millis();
    }
  } else {
    price = 0;
    display.showNumberDec(0, true);
  }
}
```

INTERMEDIATE PROJECT

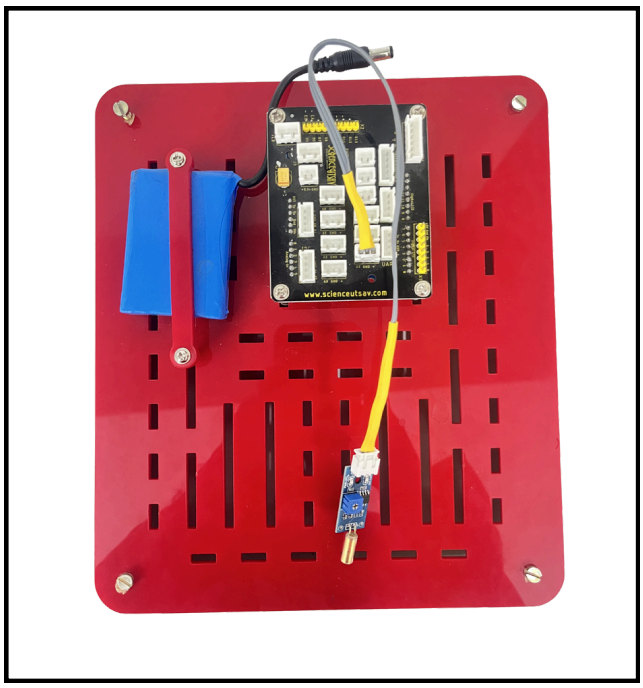
The circuit diagram of the project is given below



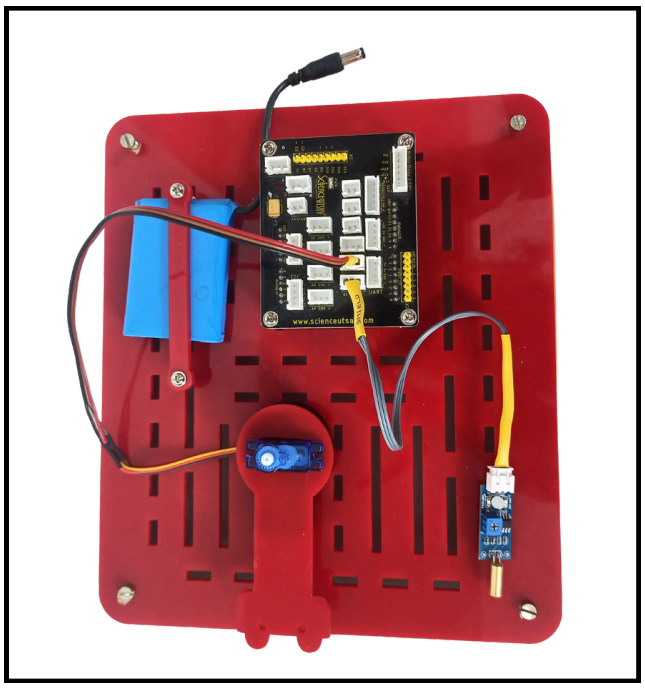
CONNECTION:

1. Tilt sensor- D2
2. Seromotor- D4

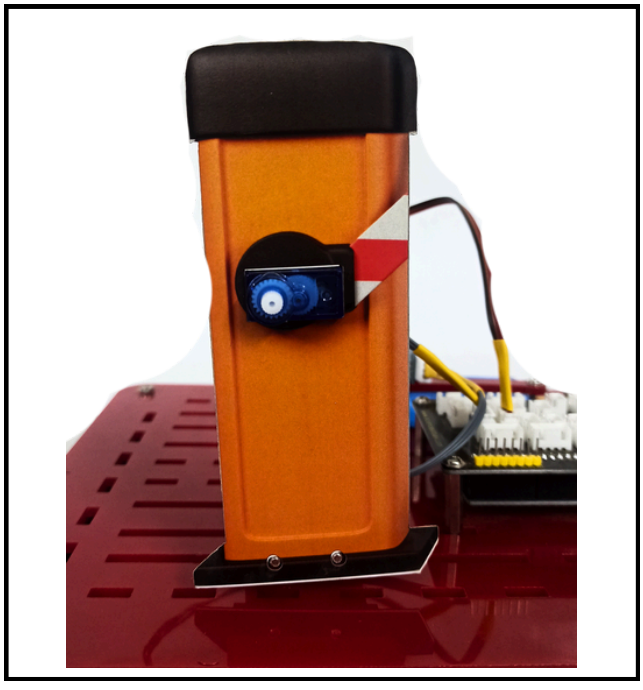
Step 1- Connect the tilt sensor to the board



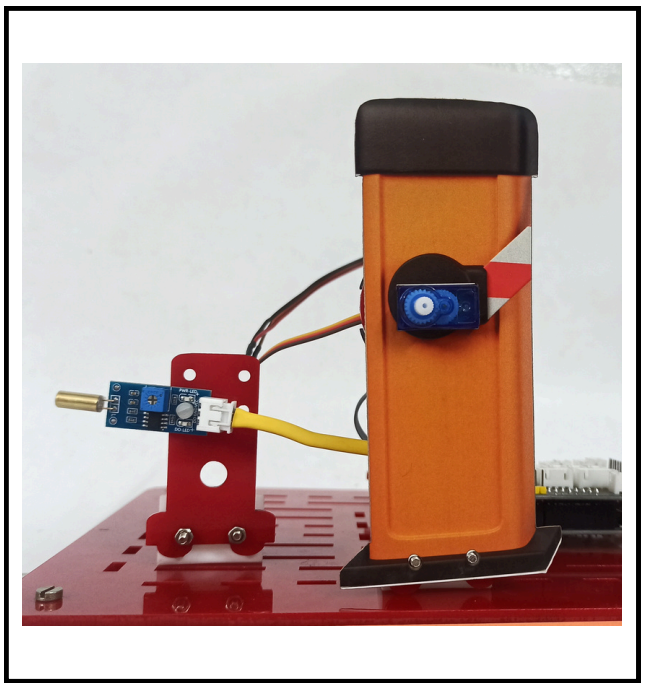
Step 2- Attach the servomotor holder as shown



Step 3- Attach the template to the stand



Step 4- Fix the tilt sensor to the other stand as shown



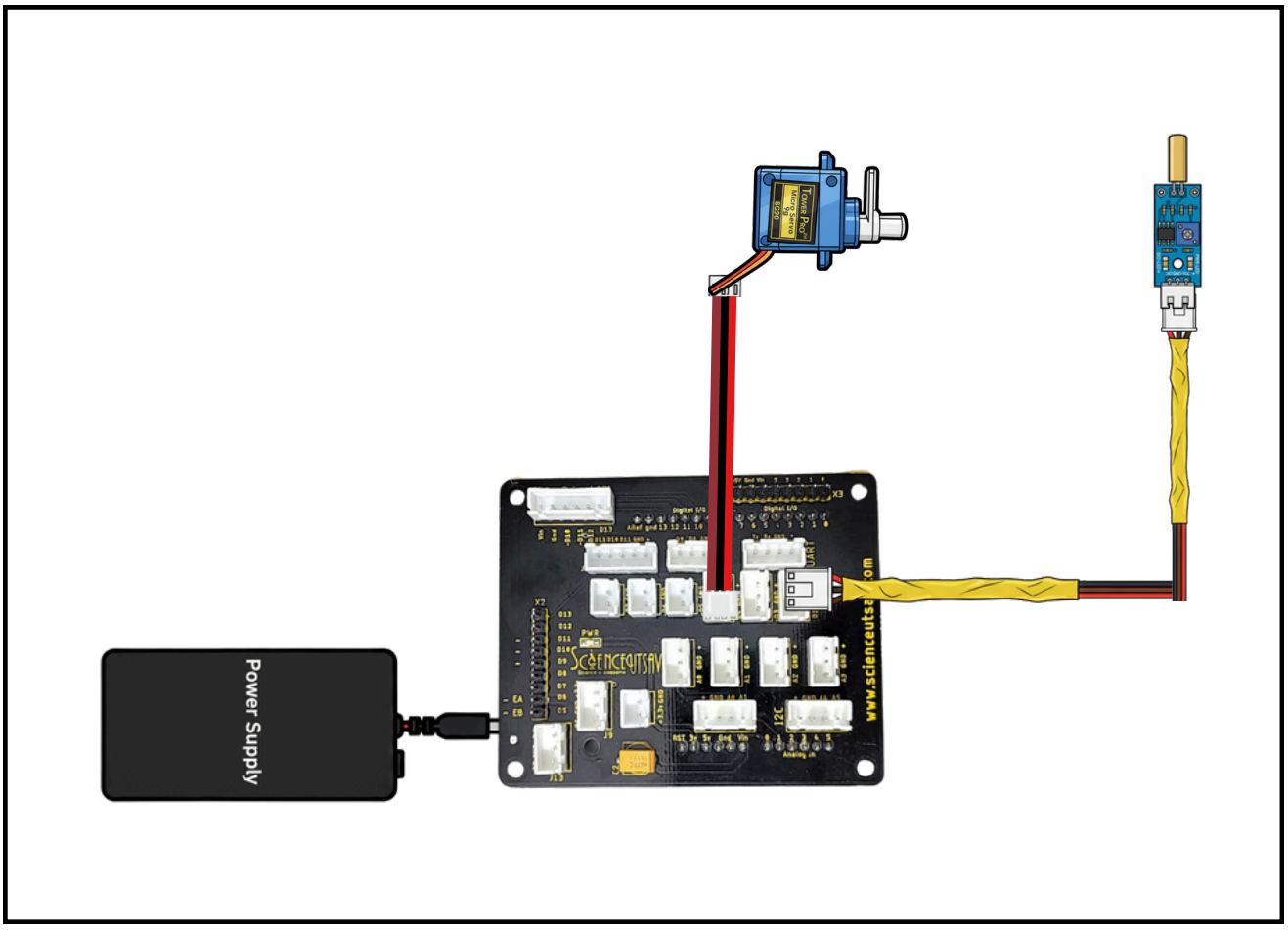
Step 5- Attach the gate template to the motor

code

```
// Session 3 Grade 7
// Taxi Gate / Barrier
// Tilt down = gate opens

#include <Servo.h>

const int tiltSensor = 2;
const int servoPin = 4;

const int TILT_ACTIVE = HIGH;  // change to LOW if opposite

Servo gateServo;

void setup() {
  pinMode(tiltSensor, INPUT);
  gateServo.attach(servoPin);

  gateServo.write(0);  // gate closed
}

void loop() {
  if (digitalRead(tiltSensor) == TILT_ACTIVE) {
    gateServo.write(90);  // gate open
  } else {
    gateServo.write(0);   // gate closed
  }
}
```

ADVANCE PROJECT

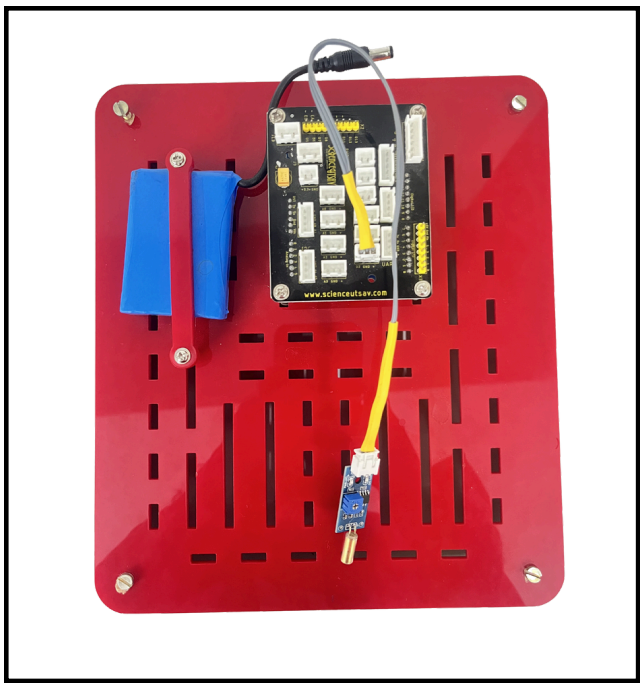
The circuit diagram of the project is given below



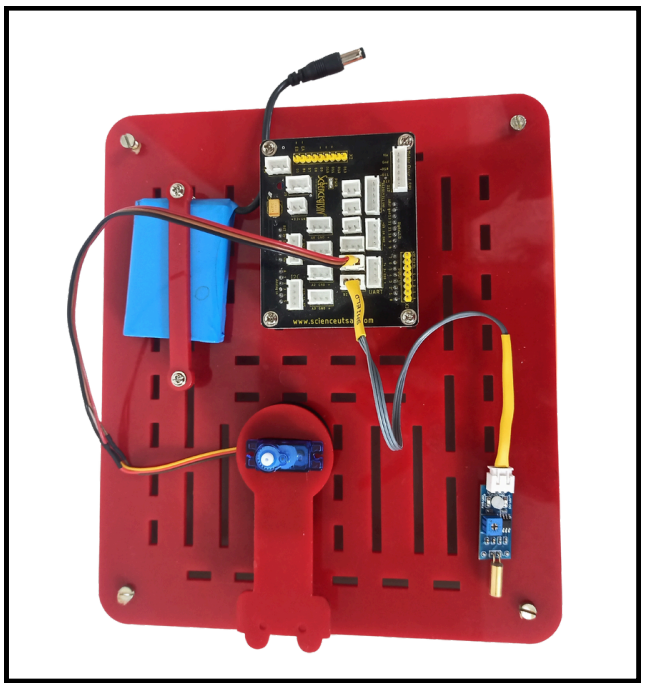
CONNECTION:

1. Tilt sensor- D2
2. Servomotor- D4

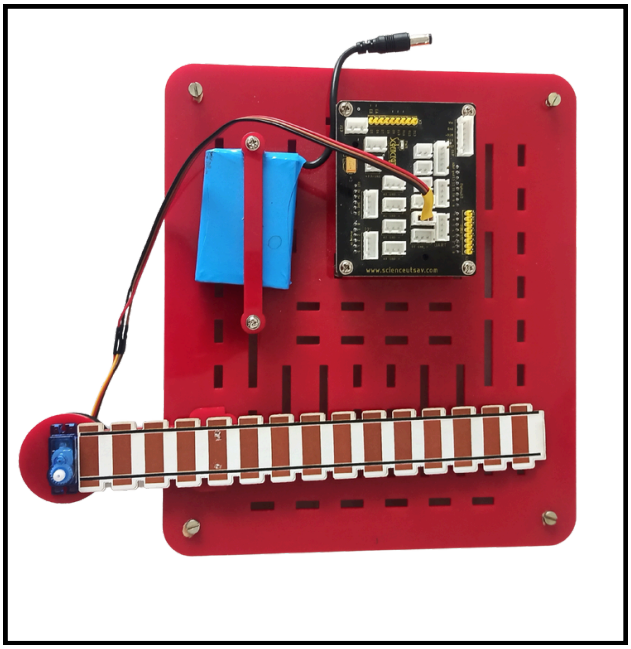
Step 1- Connect the tilt sensor to the board

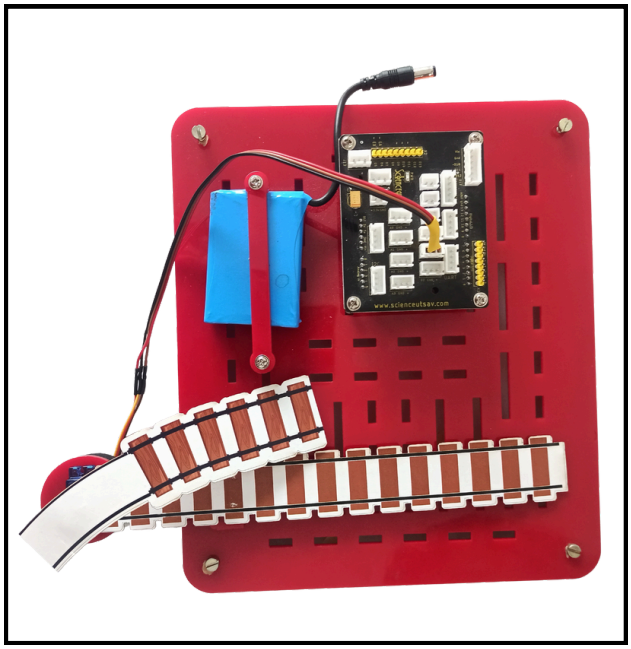


Step 2- Attach the servomotor holder as shown

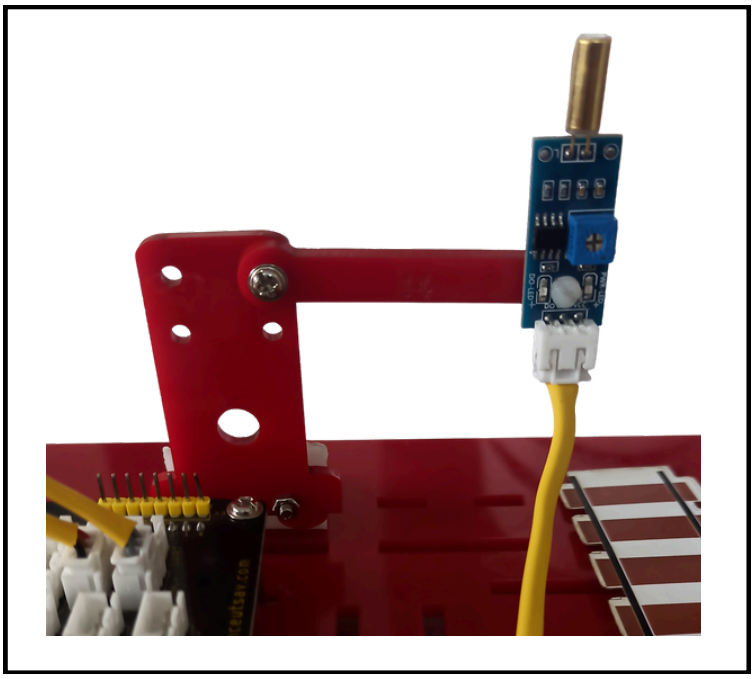


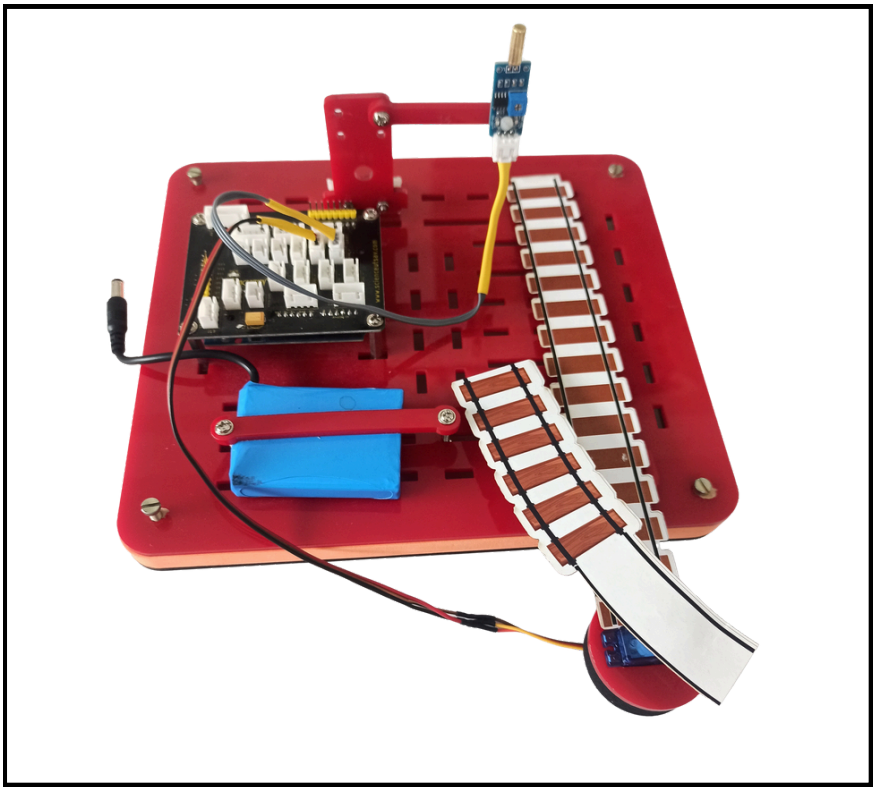
Step 3- Attach the templates as shown





Step 4- Attach the tilt sensor as shown



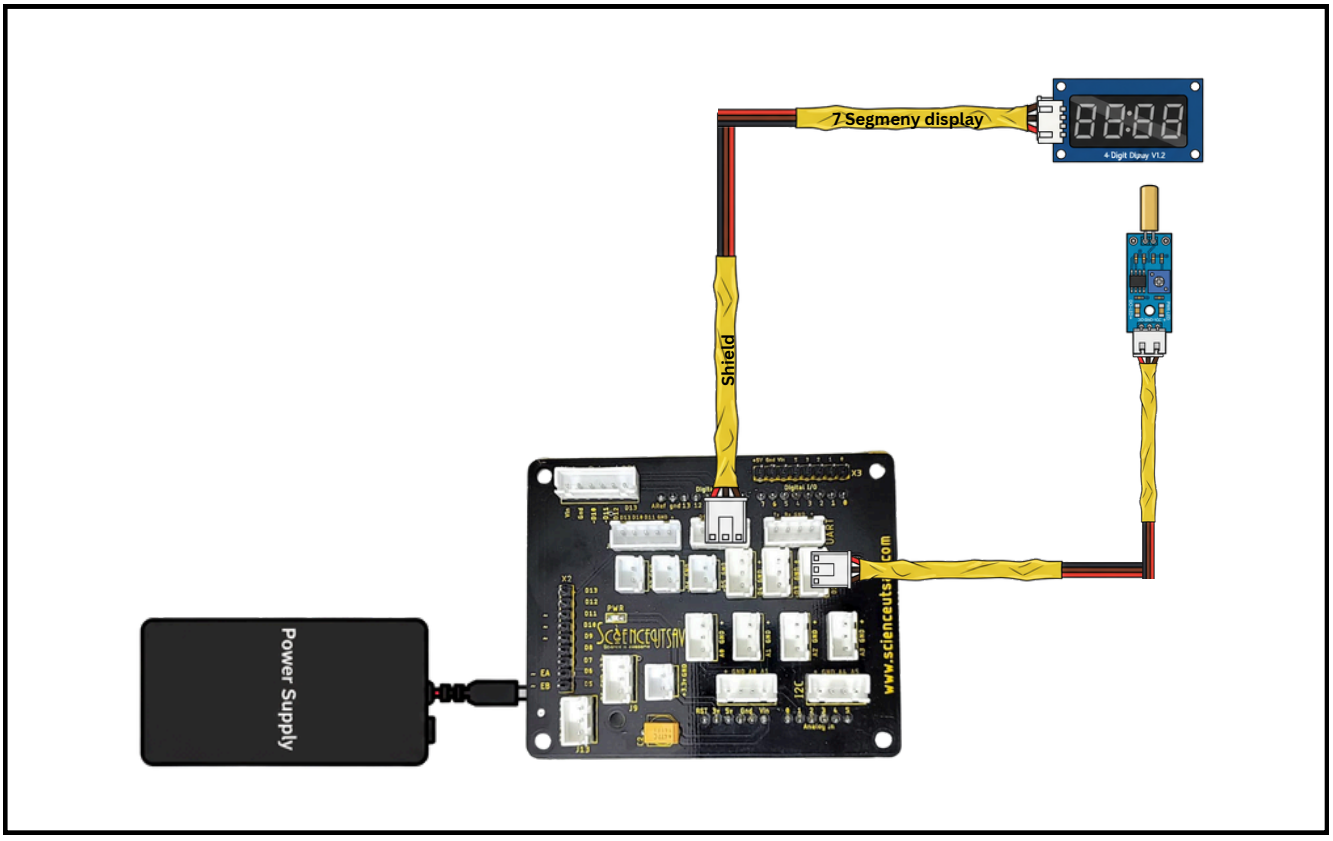


code

```
// Session 3 Grade 8
// Railway Track
// Tilt down = track moves into correct position

#include <Servo.h>

const int tiltSensor = 2;
const int servoPin = 4;

const int TILT_ACTIVE = HIGH;  // change to LOW if
opposite

Servo trackServo;

void setup() {
  pinMode(tiltSensor, INPUT);
  trackServo.attach(servoPin);

  trackServo.write(0);  // track open / wrong position
}

void loop() {
  if (digitalRead(tiltSensor) == TILT_ACTIVE) {
    trackServo.write(90);  // track correct position
  } else {
    trackServo.write(0);
  }
}
```

ADVANCE PROJECT

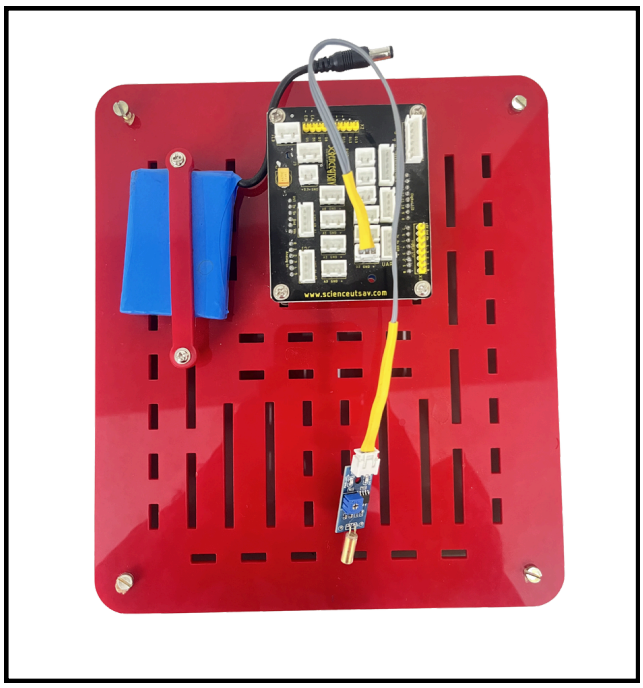
The circuit diagram of the project is given below



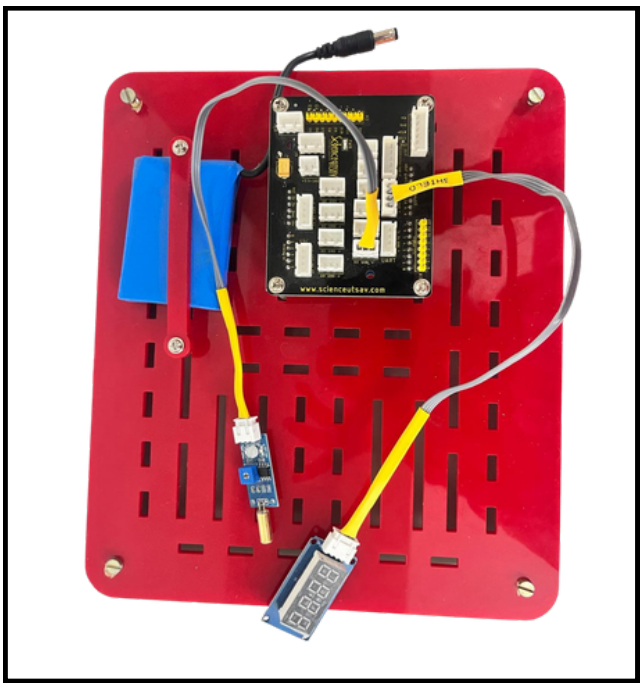
CONNECTION:

1. Tilt sensor- D2
2. 7-segment display- D8, D9

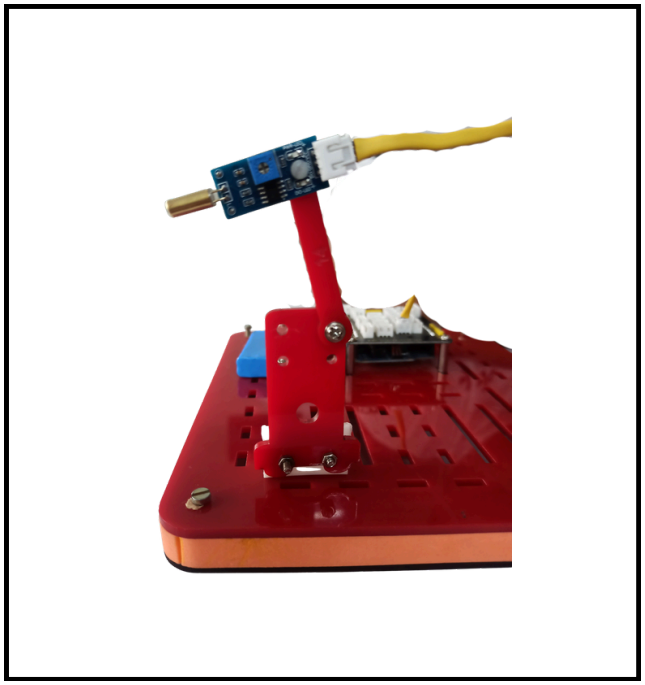
Step 1- Connect the tilt sensor to the board



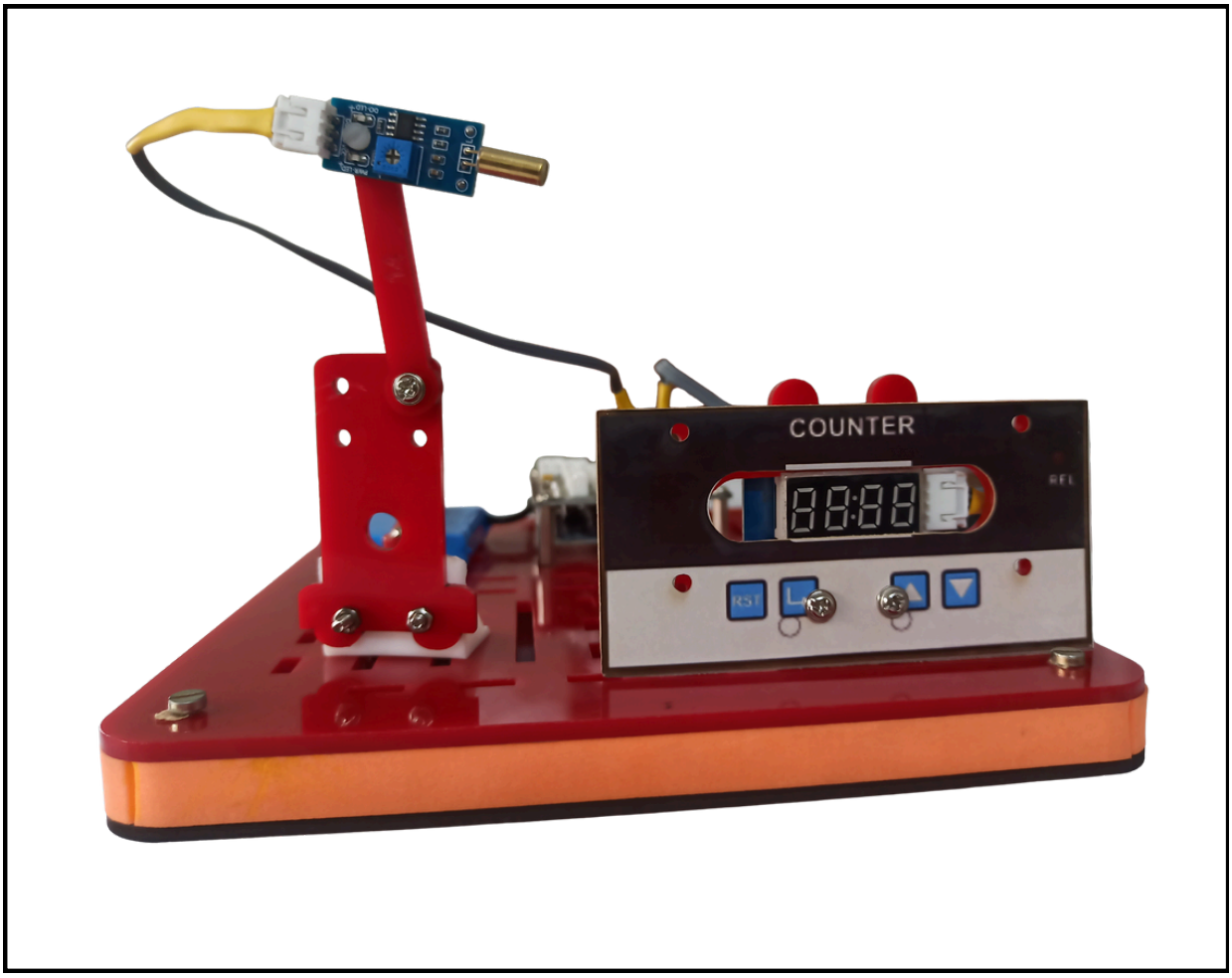
Step 2- Connect the display as shown



Step 3- Attach the sensor to the structures as shown



Step 4- Attach the counter template to L-clamp and fix it to the board

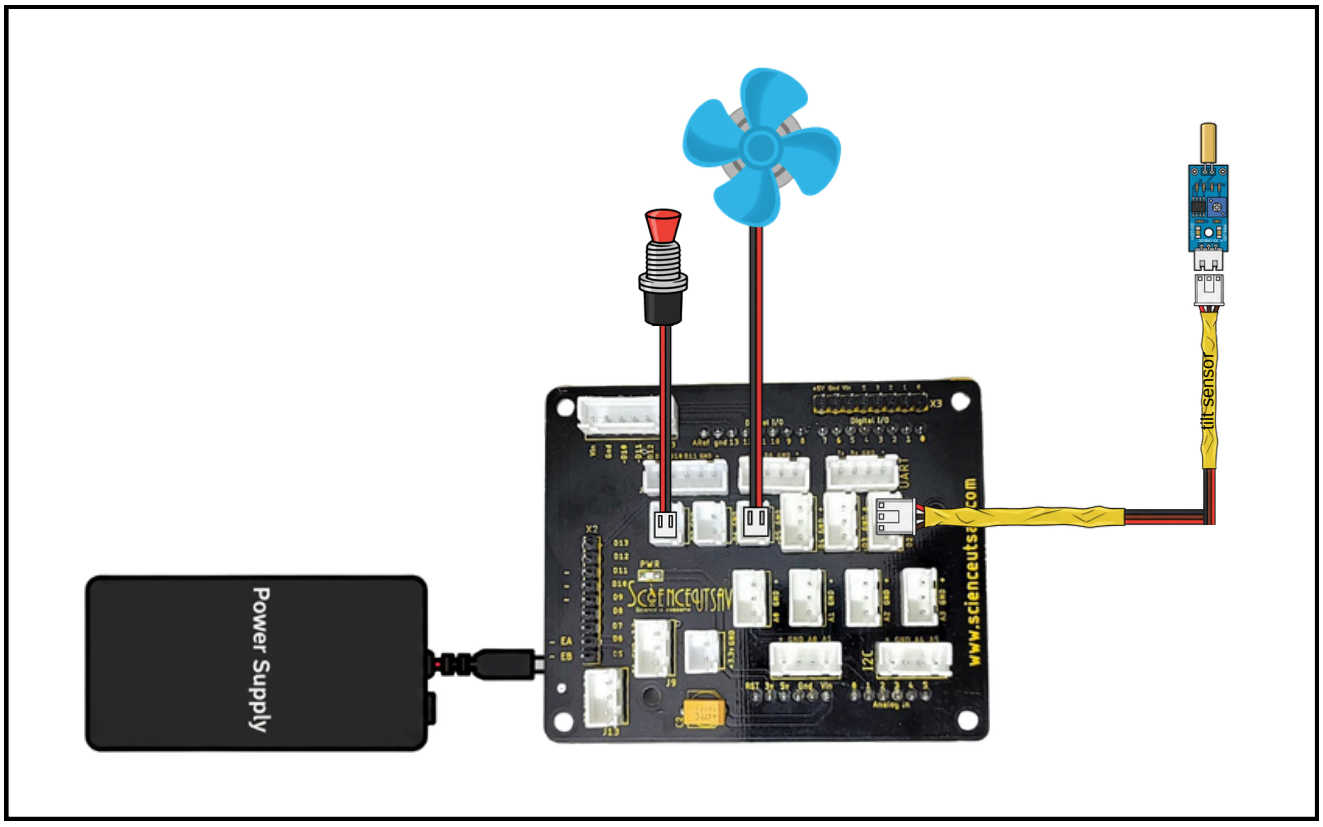

COUNTER
RST

code

```
 //Session 3 Grade 8
// Counter
// Every tilt = count increases by 1

#include <TM1637Display.h>

const int tiltSensor = 2;

const int CLK = 9;
const int DIO = 8;

const int TILT_ACTIVE = HIGH;  // change to LOW if opposite

TM1637Display display(CLK, DIO);

int countValue = 0;
bool lastTiltState = LOW;

void setup() {
  pinMode(tiltSensor, INPUT);

  display.setBrightness(7);
  display.showNumberDec(0, true);
}

void loop() {
  bool currentTilt = digitalRead(tiltSensor);

  if (currentTilt == TILT_ACTIVE && lastTiltState != TILT_ACTIVE) {
    countValue++;

    if (countValue > 9999) {
      countValue = 0;
    }

    display.showNumberDec(countValue, true);
    delay(300);
  }

  lastTiltState = currentTilt;
}
```

EXPERT PROJECT

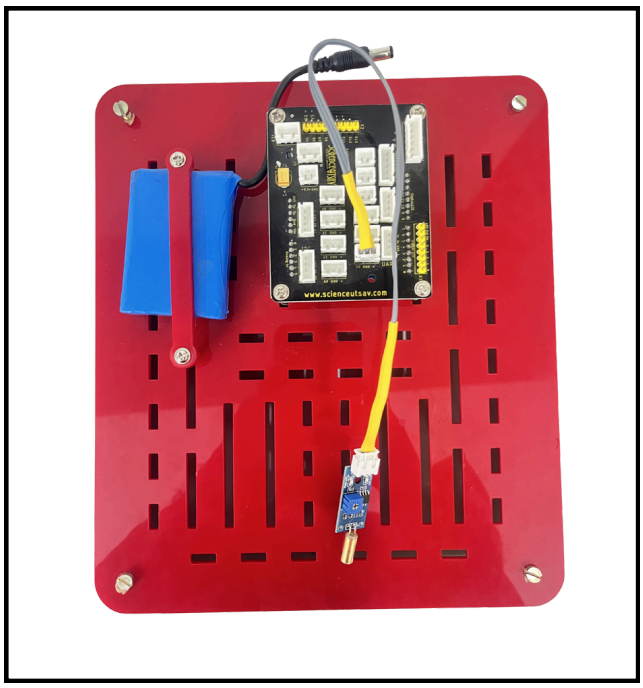
The circuit diagram of the project is given below



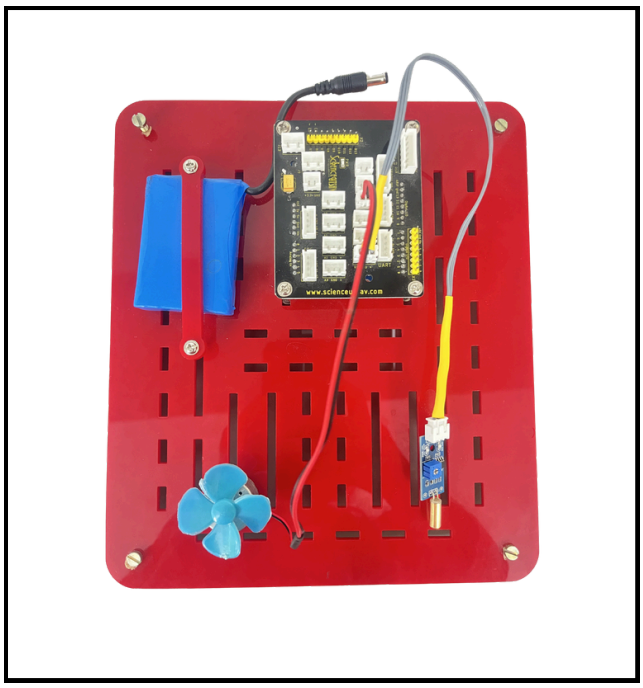
CONNECTION:

1. Tilt sensor- D2
2. DC motor- D5
3. Push button- D7

Step 1- Connect the tilt sensor to the board



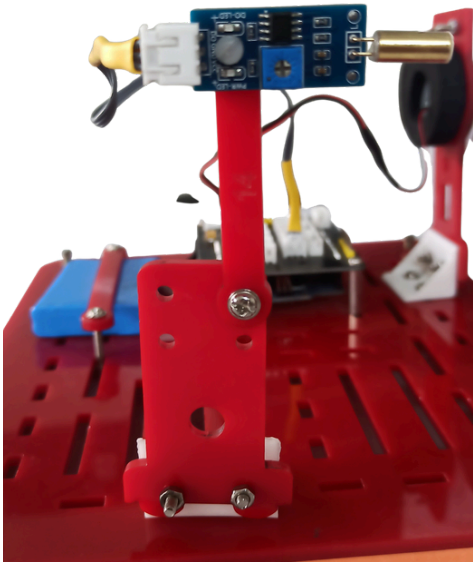
Step 2- Attach the DC motor as shown



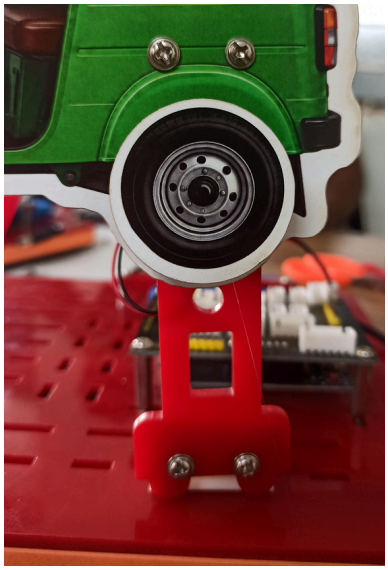
Step 3- Attach the template to the DC motor holder

Step 4- Fix the DC motor to the stand

Step 5- Fix the tilt sensor as shown



Step 6- Fix the wheel to the motor by attaching the pulley to it



Step 7- Attach the push button to the holder

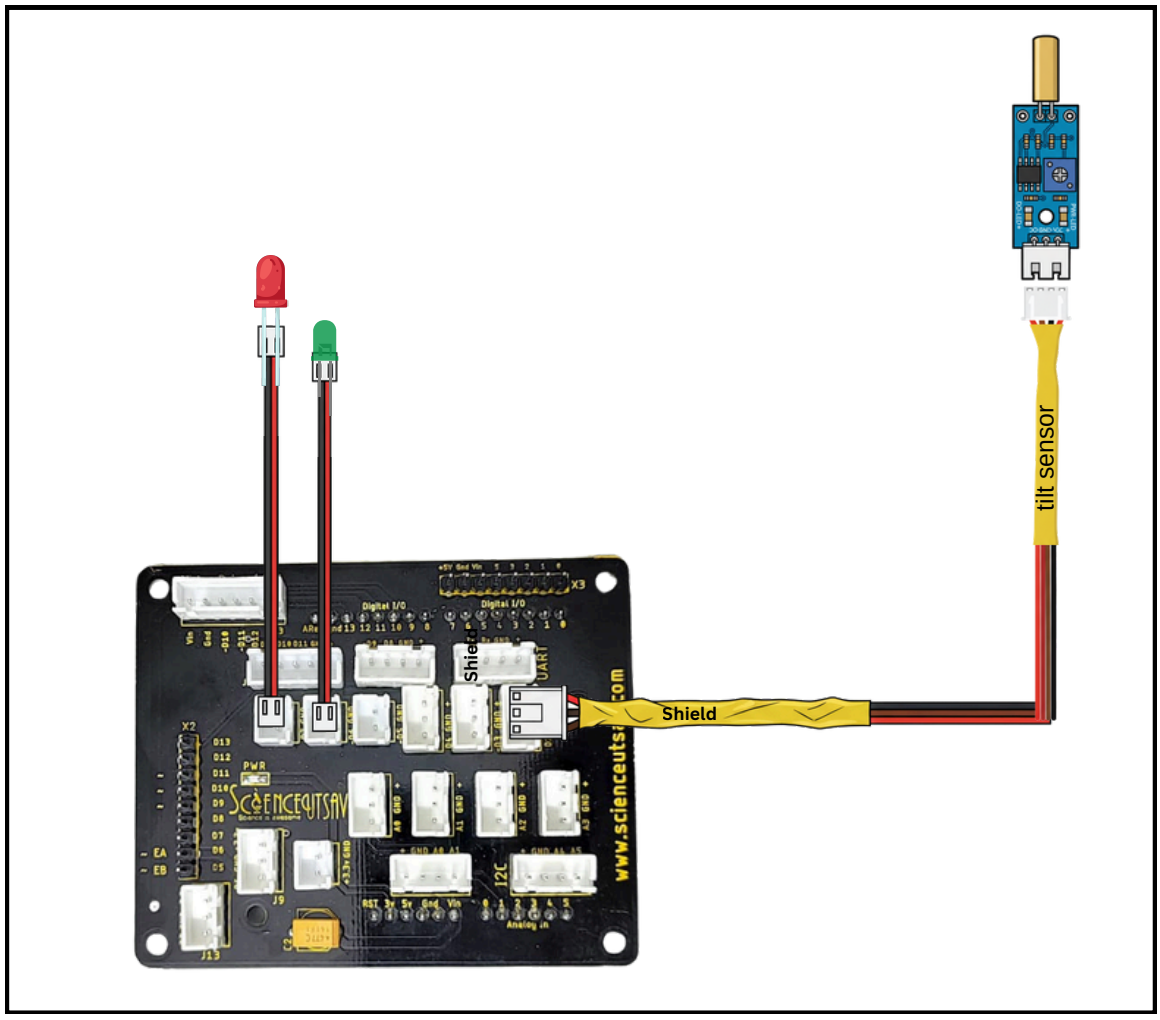
code

```
// Session 3 Grade 9
// Auto Reverse Gear
// Push button = motor forward
// Tilt + push button = motor reverse

const int tiltSensor = 2;
const int motorPin1 = 5;
const int motorPin2 = 6;
const int pushButton = 7;

const int TILT_ACTIVE = HIGH;  // change to LOW if opposite

void setup() {
  pinMode(tiltSensor, INPUT);
  pinMode(pushButton, INPUT_PULLUP);

  pinMode(motorPin1, OUTPUT);
  pinMode(motorPin2, OUTPUT);

  stopMotor();
}
```

```
void loop() {
  bool buttonPressed = digitalRead(pushButton) == LOW;
  bool tiltDetected = digitalRead(tiltSensor) == TILT_ACTIVE;

  if (buttonPressed && tiltDetected) {
    reverseMotor();
  } else if (buttonPressed && !tiltDetected) {
    forwardMotor();
  } else {
    stopMotor();
  }
}

void forwardMotor() {
  digitalWrite(motorPin1, HIGH);
  digitalWrite(motorPin2, LOW);
}

void reverseMotor() {
  digitalWrite(motorPin1, LOW);
  digitalWrite(motorPin2, HIGH);
}

void stopMotor() {
  digitalWrite(motorPin1, LOW);
  digitalWrite(motorPin2, LOW);
}
```

EXPERT PROJECT

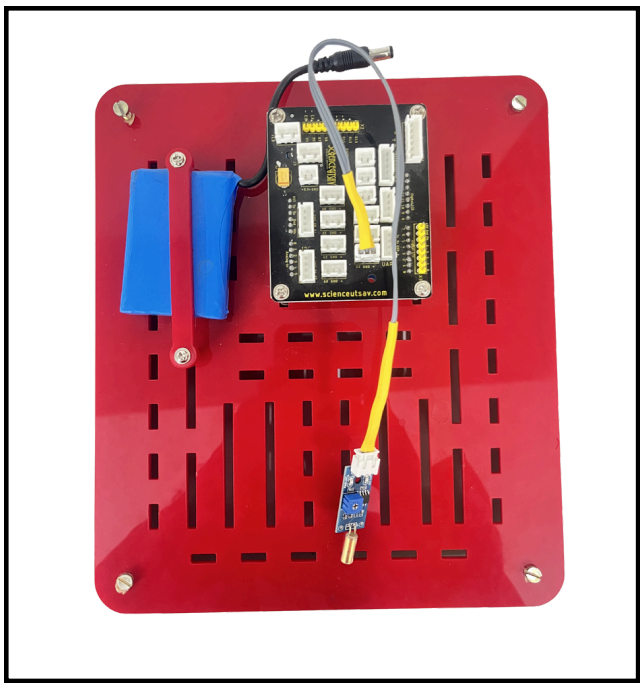
The circuit diagram of the project is given below



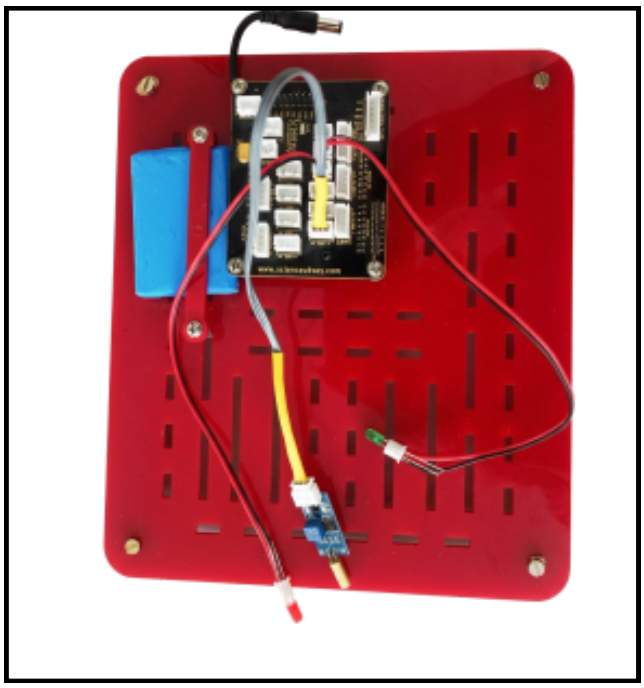
CONNECTION:

1. Tilt sensor- D2
2. Green LED- D6
3. Red LED- D7

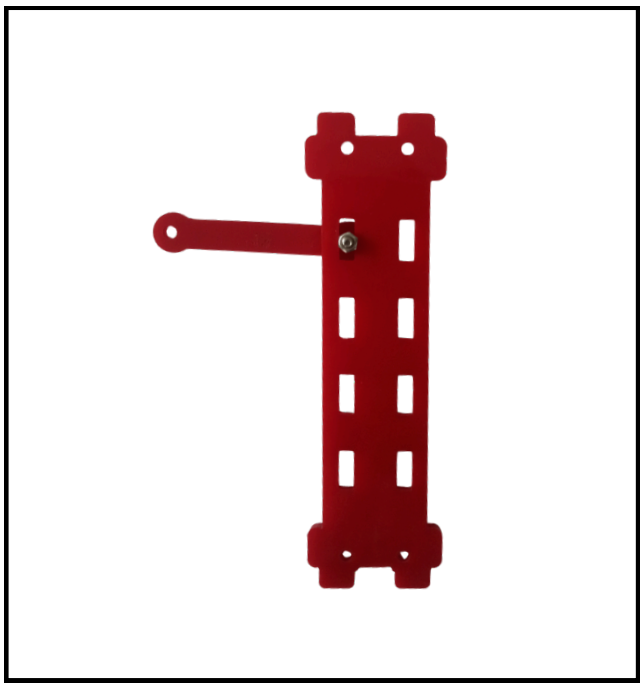
Step 1- Connect the tilt sensor to the board



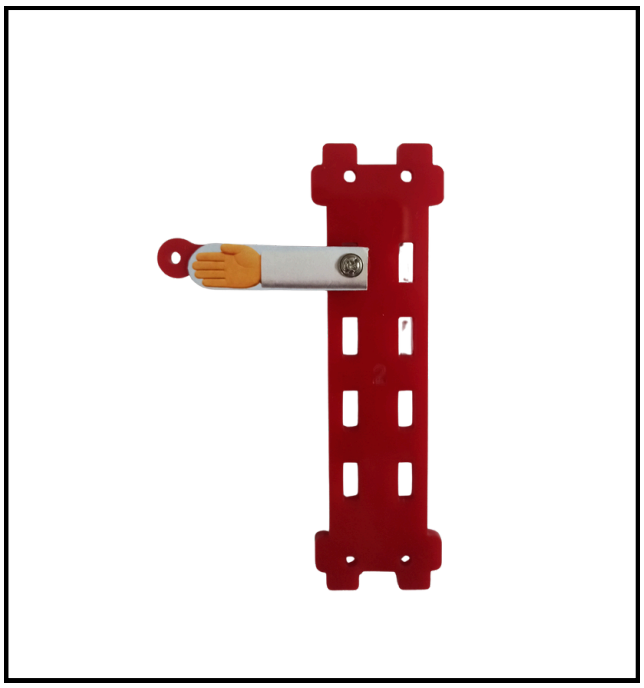
Step 2- Attach the red and green LEDs as shown



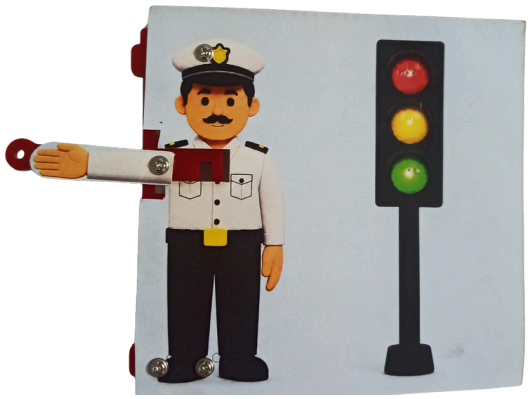
Step 3- Attach the structures as shown



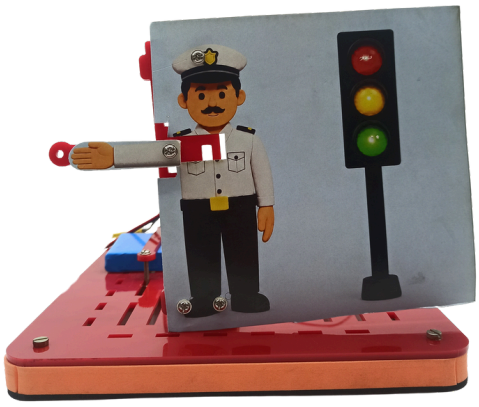
Step 4- Attach the template to the structure



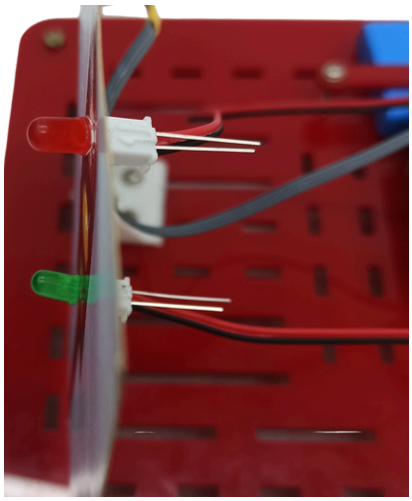
Step 5- Attach the structure to the template as show



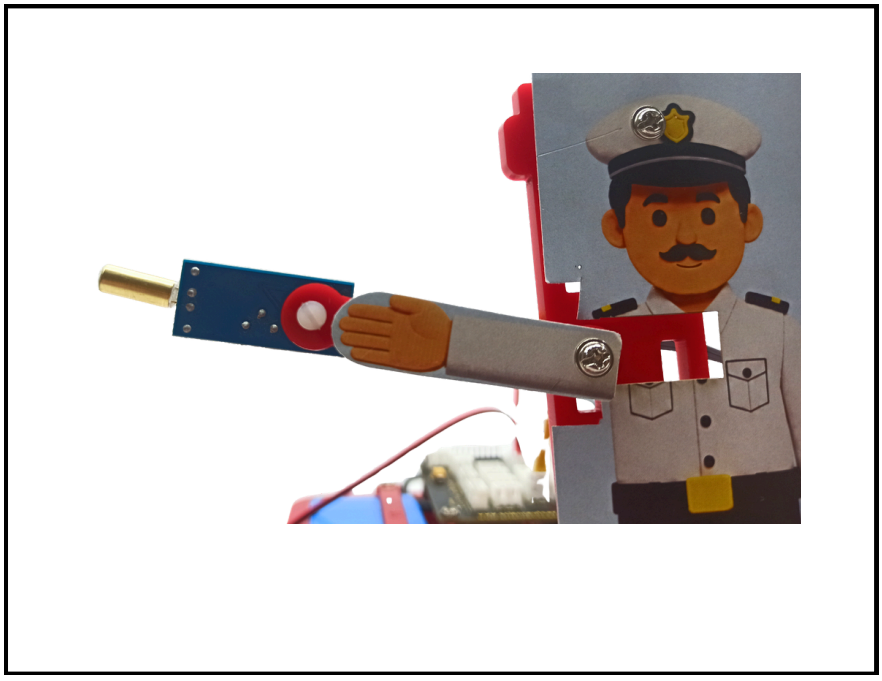
Step 6- Fix the whole thing onto the board

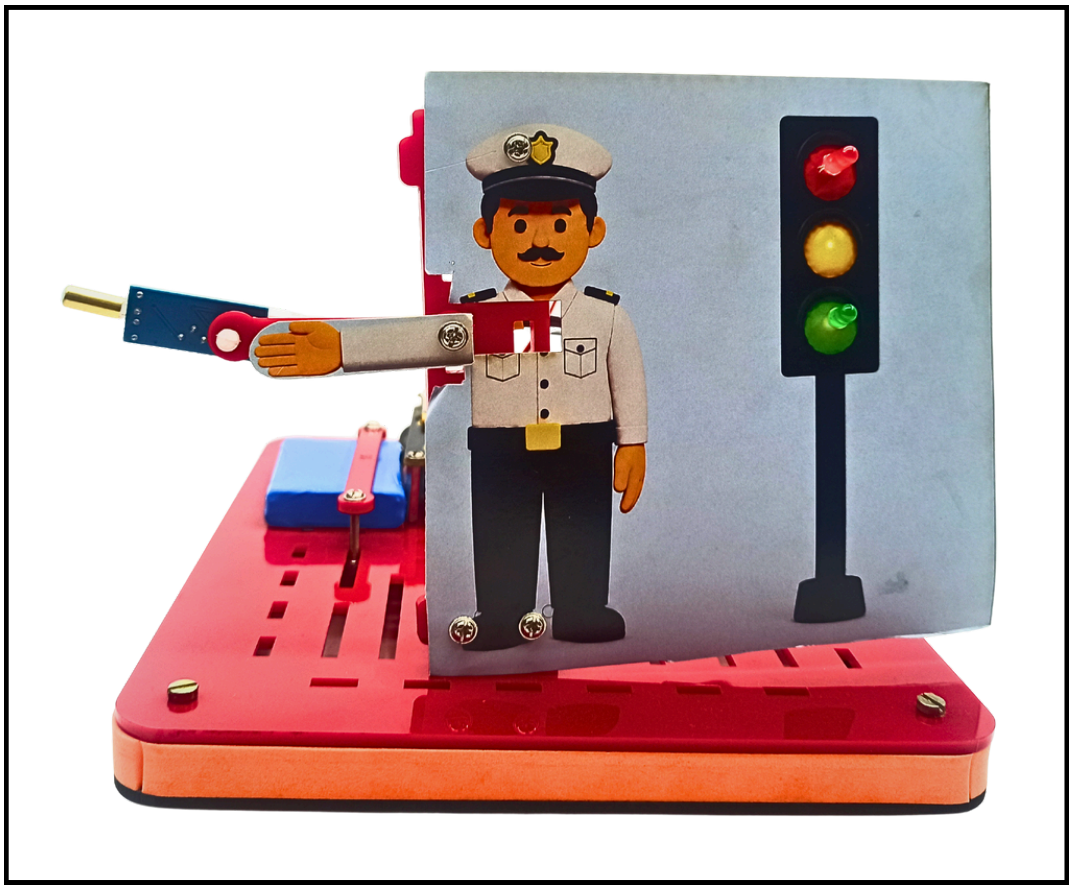


Step 7- Attach the dip LEDs to the traffic light as shown



Step 8- Attach the tilt sensor to the structure as shown





code

```
const int tiltPin = 2;
const int redLED = 7;
const int greenLED = 6;

void setup() {
  pinMode(tiltPin, INPUT_PULLUP);
  pinMode(redLED, OUTPUT);
  pinMode(greenLED, OUTPUT);
}

void loop() {
  if (digitalRead(tiltPin) == LOW) {
    // Sensor in one position (e.g., Horizontal)
    digitalWrite(redLED, HIGH);
    digitalWrite(greenLED, LOW);
  } else {
    // Sensor in the other position (e.g., Tilted/Vertical)
    digitalWrite(redLED, LOW);
    digitalWrite(greenLED, HIGH);
  }
}
```